Binomial Asset Pricing Model

Arbitrage - Risk Free Profit

TYPE I: $V_0 = 0$ (Portfolio value) $\begin{cases} P(V_T \geq 0) = 1 \\ P(V_T > 0) > 0 \end{cases}$

TYPE II: $V_0 < 0$ $\quad P(V_T \geq 0) = 1$

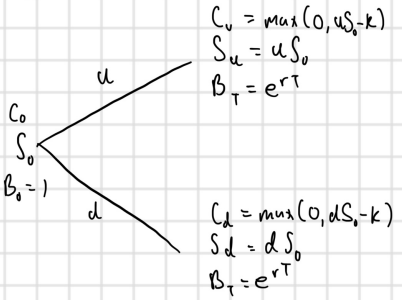
The Law of One Price:

$$P(V_T^a = V_T^b) = 1 \Rightarrow V_0^a = V_0^b$$

One period Binomial Model:

KEY
S - Stock
B - Bank Acc.
C - Derivative

Time step (T): 0 → 1



P(Value Stock, Bank T = Call option @T) = 1

⇒ Value Stock, Bank 0 = call option @0 (Via Law of one price)

⇒ $V_0 = C_0$

$V_T = \Theta \cdot (S_T, B_T) \qquad C_T$

$\Theta = \begin{pmatrix} \beta \\ \alpha \end{pmatrix}$ β = units of stock, α = units of bank account

$* \quad C_0 = \alpha B_0 + \beta S_0$

eq1) $C_u = \alpha e^{rT} + \beta S_u$

eq2) $C_d = \alpha e^{rT} + \beta S_d$

$\therefore C_u - C_d = \beta(S_u - S_d)$

$\Rightarrow \beta = \frac{C_u - C_d}{S_u - S_d}$

$\Rightarrow \alpha = e^{-rT}(C_u - \beta S_u)$

$\alpha = e^{-rT}(C_u - \beta S_u) = e^{-rT}(C_d - \beta S_d)$

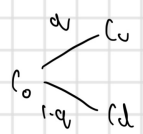
let $B_0 = 1$;

from *, $C_0 = e^{-rT}[C_d - \beta S_d + \beta S_0 e^{rT}]$

$= e^{-rT}\left[C_d + \left(\frac{C_u - C_d}{S_u - S_d}\right)(S_0 e^{rT} - S_d)\right]$

let $q = \frac{S_0 e^{rT} - S_d}{S_u - S_d}$; $\quad 0 < q < 1$

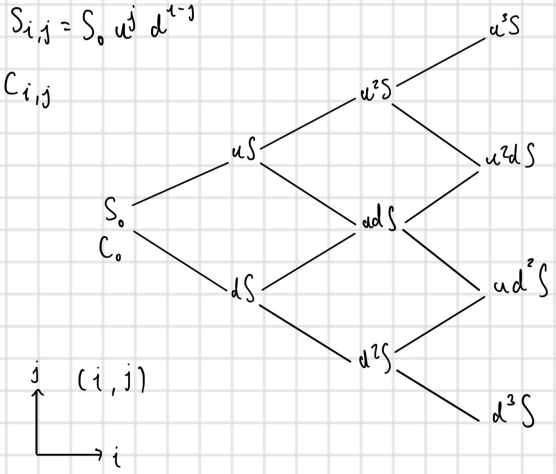
$C_0 = e^{-rT}[qC_u + (1-q)C_d]$



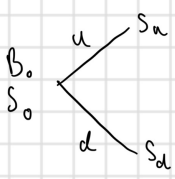
T: 0 1 2 3

$S_{i,j} = S_0 u^j d^{i-j}$

$C_{i,j}$



Arbitrage Free Pricing: $0 < d < 1+r < u$



$d > 1+r \Rightarrow$ Type I arbitrage

$u \leq 1+r \Rightarrow$ Type I arbitrage